
INTRODUCTION TO COST-BENEFIT ANALYSIS

THIS SECTION PRESENTS THE COST-BENEFIT ANALYSIS (CBA) FOR THE PROPOSED SMART HOSTEL MANAGEMENT WEBSITE FOR UNIVERSITI TECHNOLOGY MALAYSIA. THE AIM OF THE THIS ANALYSIS IS TO COMPARE THE ESTIMATED DEVELOPMENT AND OPERATIONAL COSTS OF THE SYSTEM WITH THE EXCEPTED BENEFITS FROM THE PROJECT .IT HELPS DECISION-MAKERS DETERMINE WHETHER A PROJECT IS FINANCIALLY AND PRACTICALLY WORTHWHILE. BY ANALYZING BOTH THE COSTS AND BENEFITS, THE UNIVERSITY CAN DETERMINE WHETHER IMPLEMENTING THIS DIGITAL HOSTEL MANAGEMENT SYSTEM IS A PRACTICAL AND VALUABLE SOLUTION.





The importance of feasibility studies

A feasibility study is one of the most important steps before starting any project. It is conducted to evaluate whether a project idea is practical, achievable, and likely to succeed. This study helps project owners and organizations understand the potential risks, costs, and benefits before investing time, effort, and financial resources. By conducting a feasibility study, decision-makers can make informed choices rather than relying on assumptions or uncertain expectations

Furthermore, a feasibility study helps organizations understand the economic value of the project over time. Financial techniques such as Present Value (PV), Cost–Benefit Analysis (CBA), and Profitability Index (PI) are often used to determine whether the project will produce a positive financial return. These methods allow analysts to measure the value of future income and compare it with the investment required to implement the system



Operational costs are the ongoing expenses needed to operate and maintain a system after it is developed. They ensure the system continues to run smoothly and effectively

Operational Costs [Monthly]

Item	Monthly Cost [RM]
IT Administrator /System Manager	2500RM
Cloud service	1250RM
Internet Service	RM200
System Maintenances	RM400
System Security [Fire wall] [Data protection]	RM250

Development Costs



Development costs are the initial expenses required to create and implement a system before it becomes operational. These costs occur during the early stages of the project and are necessary to prepare the system for use

Development Costs

<i>Item</i>	<i>Cost [RM]</i>
Hardware(computers,Booking system)	RM10000
Software Development(hotel booking system design)	15000RM
Advertisement &Marketing [opening promotion]	RM5000
Database Development	RM 3000

Expected Benefits

Benefit	Amount
Online Booking service fee	per month 9500M
Additional Service	3500 RM per month

This part is about realism because excessive idealism in calculations is a serious mistake in feasibility studies



Assumptions

.Discount rate 8%

.Sensitivity factor (cost) 1.1

.Sensitivity factor (benefit) 0.8

.Annual increment (cost) 5%

.Annual increment (benefit) 4%

Costs	Year 0	Year 1	Year 2	Year 3
Development Costs :				
Hardware(compute rs,Booking system)	RM10000			
Software Development(hotel booking system design)	15000RM			
Advertisement &Marketing [opening promotion]	RM5000			
Database Development	RM3000			
TOTAL	33000			
Production Costs :				
System Manager	-	R2750M	R2887M	3031 RM
Cloud service	-	RM1375	1443 RM	1515 RM
Internet Service	-	RM220	RM231	242 RM
System Maintenances	-	RM440	RM462	485 RM
System Security	-	RM275	RM288	302 RM
Annual production costs :		60720 RM	RM 63756	6694 RM 3

Present Value :		R56222 M	54679 RM	5317 RM 1
Accumulated Costs		56222 RM	110901 RM	1640 72 RM

Benefits :	Year 0	Year 1	Year 2	Year3
Online Booking service fee		91200 RM	94848 RM	98641 RM
Additional Service		33600 RM	34944 RM	36314 RM
Annual Total Benfeit		124800 RM	129792 RM	134955 RM
Present Value		115555 RM	111313 RM	107192 RM
Accumulated Benefit		115555 RM	226868 RM	334060 RM
Gain or Loss		59333 RM	115967 RM	169988 RM

**Profitable 1.69
Index
(PI)**

**Since the PI is 1.69
the project is
financially successful**



CBA
